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Experimental study and thermodynamic assessment of the erbium–hydrogen binary system

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ABSTRACT

The erbium–hydrogen (Er–H) binary system has been investigated experimentally. New solubility limits and extensions of the homogeneity domains have been measured, using several experimental techniques, and high purity materials. A thermodynamic assessment of the system using the Calphad method has been performed. The calculated phase diagram shows a fair agreement with the experimental data. Both experimental and calculated phase diagrams obtained differ significantly from the one available in the literature.

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1. Introduction

In order to increase the cycle length and the fuel burn up of light water reactors, new concepts of claddings are being investigated. One of them consists in inserting solid burnable poisons, such as gadolinium or erbium, inside the cladding [1,2]. The relevant solution studied uses an internal layer made of zirconium alloy enriched in erbium. It is inserted between two regular zirconium base alloys layers, the whole set up constituting the cladding tube. In pressurized water reactors, the cladding is surrounded by pressurized water (350 °C, 155 bars). These conditions promote the dissociation of H_2O molecule and a part of the hydrogen released diffuses inside the cladding. Thus, it is of high interest to know the interactions between hydrogen and the alloying elements inside the cladding, and more precisely the Er–Zr inner layer. Thus, the ternary system Er–Zr–H has to be studied experimentally and thermodynamically. To do so, the three associated binaries have to be well known and assessed. H–Zr and Er–Zr have already been assessed previously [3,4]. Considering the lack of experimental data, for the third binary system, Er–H, it has been studied experimentally and then optimized using the Calphad method.

2. Literature survey

The Er–H binary system reported in the literature was drawn by Mulford [5] in 1958. This phase diagram is constituted by H_2 gas and three solid phases. These solid phases include one solid solution α_{Er}, a di-hydride β-ErH_2 and a tri-hydride γ-ErH_3. The crystal structure type of these three phases is, respectively, Mg, CaF_2 and HoD_3. It can be noticed that both hydrides are non-stoichiometric. Mulford, in the absence of detailed experimental measurements of the Er–H binary system presents a rough evaluation mostly based on a supposed similarity with the Dy–H binary system. Since Mulford evaluation, some measurements have been made and reported in the literature. The crystal structures of the different hydrides of the system have been investigated by XRD by Pebler [6], Azarkh [7], Lundin [8] and Bonnet [9,10]. Neutron diffraction measurements have been made by Mansmann [11] and Udovic [12–14] to determine precisely the γ-ErH_3 crystal structure. All the crystal structures of the phases and the references can be found in Table 1.

The $\alpha_{Er}+\beta$-ErH_2 two phase region has been studied at 12 °C by Heckman [15], Singh [16] and Surplice [17] in order to determine the solubility limits in both phases. Other authors have determined the composition domains of the phases at room temperature like Pebler [6], Azarkh [7], Beavis [18] and at 325 °C like Mintz [19].

In the literature, some Pressure–Composition–Temperature (PCT) curves can be found. The PCT curves have been measured

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Table 1
Crystal structure of the phases of the Er–H binary system.

Data	α_{Er}	β-ErH_2	γ-ErH_3
Space group	$P6_3/mmc$	$Fm\overline{3}m$	$P\overline{3}c1$
Phase prototype	Mg	CaF_2	HoH_3
Lattice parameters	a=3.559 Å c=5.587 Å	a=5.123 Å	a=6.272 Å c=6.526 Å
Hydrogen location	Tetrahedral sites then octahedral sites	Tetrahedral sites occupied[a]	Tetrahedral sites, 1/3 of triangular sites and 2/3 of displaced octahedral sites occupied
References	[37]	[6–8,10]	[6,8,11]

[a] Over stoichiometry accommodated by additional hydrogen in octahedral sites.

Table 2
List of the experimental data of the literature.

Type of data	Composition range	Temperature range (°C)	Reference	Data used for modelling
Invariant	GAS + HCP_A3 → FCC_C1	1221	[8]	Yes
	GAS + FCC_C1 → HCP_A3	383	[8]	Yes
Solubility limits	α_{Er}–β-ErH_2	12	[15–17]	Yes [15] No [16,17]
	α_{Er}–β-ErH_2	25	[6,7,18]	Yes [6,7] No [18]
	α_{Er}–β-ErH_2	350	This work	Partly Yes
	α_{Er}–β-ErH_2	$24 < T < 851$	[21]	Yes
	α_{Er}–β-ErH_2	$500 < T < 950$	[8]	Yes
	α_{Er}–β-ErH_2	$700 < T < 900$	[22]	No
	β-ErH_2–γ-ErH_3	12	[38]	Yes [15,16] No [17]
	β-ErH_2–γ-ErH_3	25	[6,7]	Yes
	β-ErH_2–γ-ErH_3	$200 < T < 350$	[8]	Yes
	β-ErH_2–γ-ErH_3	325	[19]	Yes
	β-ErH_2–γ-ErH_3	350	This work	No
Plateau pressure	α_{Er}–β-ErH_2	$500 < T < 950$	[8]	Yes
	α_{Er}–β-ErH_2	$900 < T < 1000$	This work	Yes
Solubility branches	β-ErH_2–γ-ErH_3	$200 < T < 370$	[8]	Yes
	β-ErH_2–γ-ErH_3	$350 < T < 500$	This work	Yes
	α_{Er}	$600 < T < 950$	[8]	Yes
	α_{Er}	$900 < T < 1000$	This work	Yes
	β-ErH_2	$900 < T < 1000$	This work	Yes

by Lundin [8] between the erbium solid solution α_{Er} and the di-hydride β-ErH_2 in the temperature range 500–950 °C. Lundin also measured some PCT curves between the two hydrides (β-ErH_2 and γ-ErH_3) from 200 °C to 370 °C. To do so, a hydrogenation bench (Sieverts apparatus) has been used with a silica tube as sample container. This silica tube limits the working pressure up to 1 bar. Lundin also did these experiments on the Er–D binary system [20], in which the results were similar. Several measurements were also made by Beaudry using an equilibration analytical technique [21]. These measurements were conducted on the α_{Er} and the two phase region (α_{Er} + β-ErH_2) based on several annealing treatments between 250 °C and 850 °C. Beaudry also made some XRD measurement at ambient temperature. Chernikov et al. [22] measured a single pressure–composition curve at 800 °C which is in agreement with the measurements of Lundin [13]. All the phase diagram data available in the literature on the Er–H binary system is summarized in Table 2.

Finally, ab-initio calculations have been performed on this binary system by Yang et al. [23]. They studied the hydrogen locations in the erbium solid solution (α_{Er}) for an infinite dilution. Their calculations show that H prefers locating in the tetrahedral sites. Recent ab-initio calculations made by Joubert and Crivello presented the enthalpies of formation of each phase (stable and metastable) of this binary system [24]. The result they obtained for the *fcc* erbium lattice stability is in good agreement with the value extrapolated by Cacciamani which was based on the Er–Th binary system [25]. Also, the calculated enthalpies of formation for both stable compounds, β-ErH_2 and γ-ErH_3, are in agreement with the experimental measurements.

3. Experimental work

3.1. Synthesis of the samples

For the synthesis of the samples, distilled erbium (purity of 99.95%, China Rare Metals) were used as starting materials and pure hydrogen (99.9999%). Before the synthesis the rare earth was conserved in a glove box, in order to prevent any oxidation due to air contamination. Samples of 1 g were synthesized inside stainless steel sample holders placed in a resistance furnace. For the synthesis, we used the solid gas reaction at 350 °C in hydrogenation benches. When the equilibrium was reached, the samples were air quenched. After the synthesis, the stainless steel sample holder can be opened under ambient atmosphere due to the very high stability of the hydride phases in air and the absence of pyrophoricity [7].

3.2. Phase structure characterisation and chemical composition investigation

The samples have been characterized by XRD at room temperature and under air on a Bruker D8 Advance diffractometer in order to determine the phase equilibria and the phase structures. The XRD patterns have been processed by Rietveld refinement [26] with the Fullprof program [27], in order to refine the cell parameters, and obtain the volume fraction of each phase. For the γ-ErH_3 phase, a texture parameter has been introduced in the Rietveld refinement due to the shape of the grains. As shown in Fig. 1, this phase forms grains with triangular base prism shape.

Fig. 1. SEM micrography of a γ-ErH_3 simple phase sample.

In a Bragg Brentano experiment, these prisms tend to align parallel to the flat plane thus modifying the relative intensity of the diffraction peaks.

To measure the chemical composition of each phase, the ion beam analysis has been used [28] at the nuclear microprobe of DSM/IRAMIS/SIS2M/LEEL [29], CEA Saclay, France. For these experiments, the combination of RBS (Rutherford BackScattering) and ERDA (elastic recoil detection analysis) has been used [30]. The first method, RBS, is used to obtain the heavy elements such as Er or Au coating concentration, and ERDA gives exclusively the H content.

Before the analysis, the samples have been prepared in small brass tubes filled of resin, polished, joined together and then metalized with a gold coating of roughly 10 nm.

For the analysis, a 3 μm × 3 μm alpha beam of 3 MeV has been used. Due to the grazing angle incidence, this turns to be a 12 μm × 3 μm beam on the sample surface. Counting time was between few hours for low hydrogen content to half an hour for samples with higher hydrogen content such as γ-ErH_3.

After the analysis, the raw data have been sorted with the RISMIN software [31] in order to obtain the different spectra and cartographies. Then, the SIMNRA software [32] allows the determination of the elemental composition from the ERDA and RBS spectra.

3.3. Solubility limits by pressure–composition–temperature curve measurements

In order to complete the work done by Lundin [8], new PCT curves have been measured at higher pressure and temperature. Between the solid solution α_{Er} and the di-hydride β-ErH_2 the measurements have been made at 900 °C and 1000 °C and between the two hydrides from 350 °C to 500 °C. To do so, hydrogenation benches (Sieverts apparatus) were used with 99.9999% pure hydrogen. For the $\alpha_{Er}+\beta$-ErH_2 composition range, the erbium has been put inside a molybdenum container itself inside a silica tube into a furnace. To limit the sample holder volume, a silica rod has been introduced into the silica tube. In this case, due to the silica tube, the experiments were limited to the pressure of 1 bar maximum. For the β-$ErH_2+\gamma$-ErH_3 two phase region, the erbium was introduced inside a stainless steel sample holder, placed in a furnace, as for the sample synthesis. The pressure was measured by a pressure sensor. A plotting table has been used in order to control if the equilibrium is reached. In both cases (silica or stainless steel) the assumption made was that the Er absorption kinetic is more important than the hydrogen permeation in silica and also more important than the hydrogen diffusion in stainless steel.

3.4. Results

Six samples were synthesized in the whole composition range at 350 °C and analysed by XRD at room temperature but only five samples gave results with this technique. This is due to the impossibility of having a flat shape of one sample for the XRD analysis. All the equilibria and lattice parameters obtained are summarized in Table 3.

Table 3 also presents the results of these six samples characterized by the ERDA combined with RBS technique. For each sample, we obtain, as a result, several cartographies such as in Fig. 2. RBS cartography presents the erbium location and sample defects such as fractures can be seen. ERDA cartography presents the hydrogen location. Fig. 2 shows a two phase sample ($ErH_{1.536}$) constituted by an α_{Er} solid solution in blue and the hydride β-ErH_2 in yellow/green. After processing these data with the SIMNRA software, two spectra are obtained (Fig. 3). The RBS spectrum gives the Er contribution. The Au coating can be seen as a sharp pic at high energies. In the ERDA spectrum an important difference can be observed between the hydride region and the solid solution.

Fig. 4 presents two PCT curves measured between the solid solution of erbium and the di-hydride at 900 °C and 1000 °C. These PCT curves are compared with the PCT curves measured by Lundin in the temperature range 750–950 °C [8].

Finally, Fig. 5 presents a new set of PCT curves measured between the two hydrides β-ErH_2 and γ-ErH_3 from 350 °C to 500 °C. In this Fig. 5, our 350 °C PCT curve can be compared to the 350 °C PCT curve measured by Lundin [8].

3.5. Discussion

All the phase equilibria and crystal structures found in this study using XRD measurements followed by Rietveld refinement are consistent with the literature. The lattice parameters of all the phases obtained in our work, Table 3, are in good agreement with those previously measured in other works, Table 1.

Table 3
Lattice parameters of the different samples determined by XRD and their chemical composition measured by ERDA+RBS.

Sample composition (at%)	H/M	Phases	Volume fraction (%)	Lattice parameters *a* (Å)	*c* (Å)	Composition measured by ERDA (at%)	Composition measured by ERDA H/M	Area of the zone (%)
$Er_{58.8}H_{41.2}$	$ErH_{0.7}$	α_{Er}	n.o.	3.56	5.581	$Er_{83.9}H_{16.1}$	$ErH_{0.19}$	2.1
		β-ErH_2	n.o.	5.119		$Er_{30.0}H_{70.0}$	$ErH_{2.33}$	27.1
$Er_{48}H_{52}$	$ErH_{1.082}$	α_{Er}	n.o.	n.o.	n.o.	$Er_{92.8}H_{7.2}$	$ErH_{0.08}$	2.5
		β-ErH_2	n.o.	n.o.		$Er_{31.2}H_{68.8}$	$ErH_{2.20}$	8.9
$Er_{39.4}H_{60.6}$	$ErH_{1.536}$	α_{Er}	15.1	3.566	5.602	$Er_{60.0}H_{40.0}$	$ErH_{0.66}$	7.6
		β-ErH_2	84.9	5.125		$Er_{37.3}H_{62.7}$	$ErH_{1.68}$	23.6
$Er_{35.1}H_{64.9}$	$ErH_{1.85}$	β-ErH_2	100	5.124		$Er_{34.9}H_{65.1}$	$ErH_{1.86}$	30
$Er_{30.1}H_{69.9}$	$ErH_{2.325}$	β-ErH_2	77.9	5.122		$Er_{39.0}H_{61.0}$	$ErH_{1.56}$	2.6
		γ-ErH_3	22.1	6.268	6.525	$Er_{31.3}H_{68.7}$	$ErH_{2.19}$	4.7
$Er_{25}H_{75}$	ErH_3	γ-ErH_3	100	6.264	6.52	$Er_{27.5}H_{72.5}$	$ErH_{2.64}$	2.2

n.o.: not obtained.

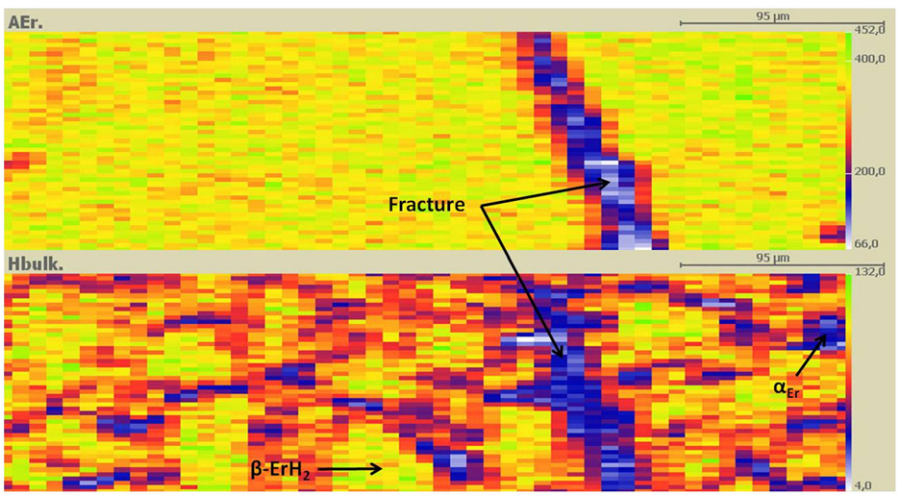

Fig. 2. Cartographies of Er and H, obtained, respectively, by RBS (on the top) and ERDA (at the bottom) cartographies of the $ErH_{1.536}$ ($Er_{39.4}H_{60.6}$) sample. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

PCT curve measurements, conducted in the α_{Er}+β-ErH_2 two phase region, are consistent with the previous survey. In fact, the 900 °C PCT curve is well between the 850 °C and the 950 °C measured by Lundin [8]. And the 1000 °C PCT curve has a higher plateau pressure than the 950 °C as expected, Fig. 4.

To avoid loss of hydrogen by diffusion in stainless steel which appears at high temperature (above 400 °C), our PCT curves had to be measured by rapid point measurements. Even if the reaction kinetic between erbium and hydrogen is high, it was compulsory verifying that measuring a PCT curve in a short time range is giving the same result than a traditional measurement. In order to check the accuracy of the method, a PCT curve has been measured at 350 °C between the two hydrides β-ErH_2 and γ-ErH_3 and then, compared to the measurements made by Lundin. This method has been validated by comparison of both measurements at 350 °C, as shown by Fig. 5. In Fig. 5, the β-ErH_2 solubility branches are in good agreement, so do the plateau pressures. The solubility branches in the γ-ErH_3 solid solution are the best that could be measured. Due to hydrogen diffusion in stainless steel, the final values of hydrogen content might be overestimated.

During ERDA combined with RBS analysis the conditions for the samples are severe: pressure of 10^{-6} mbar under a 3 MeV ion beam. In order to check if there is hydrogen desorption from the samples, they have been analysed by μXRD before and after ERDA combined with RBS analysis. For this μXRD study, a grain which has been analysed by the alpha beam has been chosen. The samples studied with these analysis where pure β-ErH_2 and pure γ-ErH_3. The γ-ErH_3 phase has a very small composition domain and a small loss of hydrogen leads to the β-ErH_2+γ-ErH_3 two phase region where a change of crystal structure appears. Since μXRD patterns of the sample before and after analysis by ERDA combined with RBS are very similar for both phases, Fig. 6, it confirms that there is no desorption during analysis. The main difference between the patterns is due to the grain texture: only one grain is analysed and it is sealed in the resin which allows any crystal orientation. This can be easily seen after the treatment of the patterns which shows that the crystal structure is the same before and after the analysis for both samples.

The spatial resolution of ERDA combined with RBS analysis (3 µm × 12 µm) is not sufficient to resolve small precipitates or fine microstructure of solid solutions. No valid chemical composition can be obtained for these individual small areas. This explains some uncertainties from Table 3. Also, the samples are sealed in the transparent resin which has a higher content of hydrogen than the solid solution but a smaller content than the hydrides. Due to the thin grain structure of the γ-ErH_3 powder, some resin might be between the grains. This can explain the difficulty we had to measure the right hydrogen content in this phase and the lower value actually measured compared to the expected one.

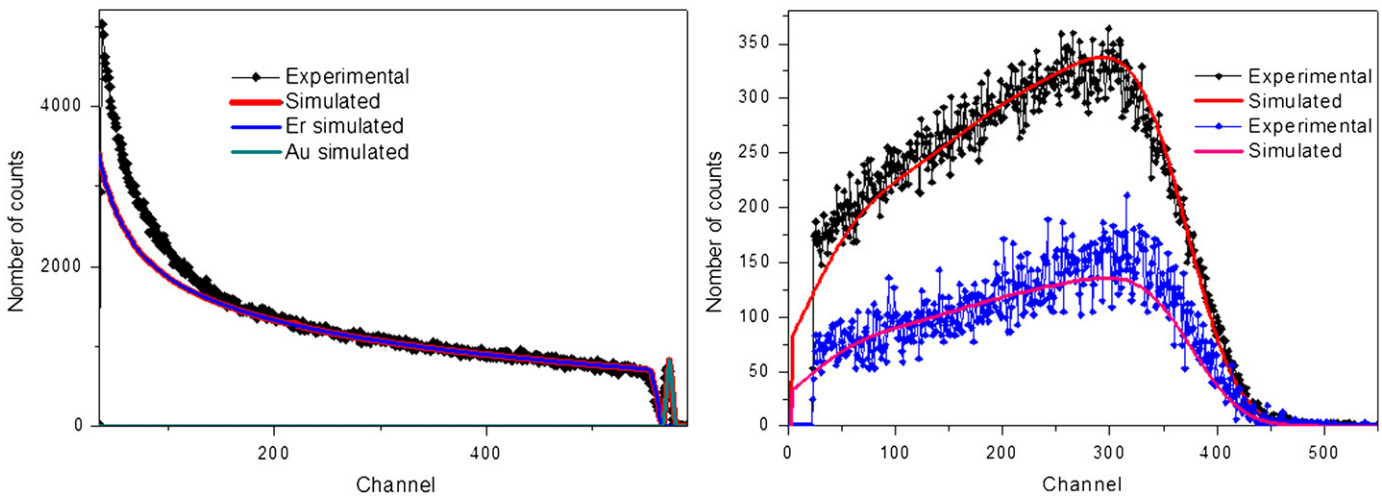

Fig. 3. RBS (on the left) and ERDA spectra (on the right) of the $ErH_{1.536}$ sample. On the ERDA spectrum, the black points belong to the hydride zones and the blue points are those of the solid solution. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

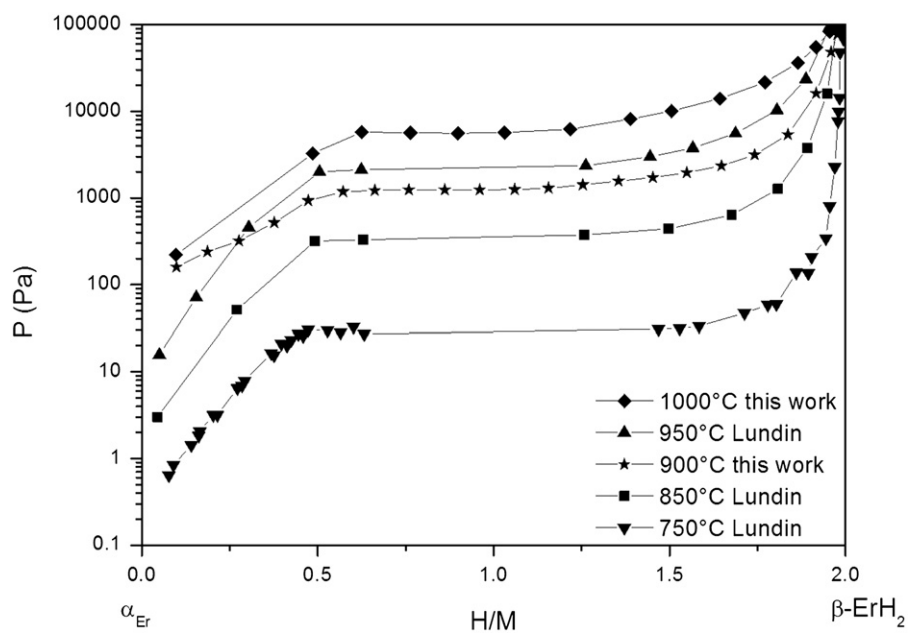

Fig. 4. PCT curves measured at 900 °C and 1000 °C between the solid solution and the di-hydride (α_{Er} and β-ErH_2) compared to Lundin's measurements from 750 °C to 950 °C [8].

4. Thermodynamic modelling

The Er–H binary phase diagram has been assessed with the Calphad method using the Thermo-Calc software [33]. This technique consists in describing the Gibbs energies of all the phases present in the system (stable and metastable) as a function of temperature and composition. Due to the absence of any experimental data on the liquid phase, this phase has not been considered.

4.1. Pure elements

The Gibbs energies of the pure elements are defined with respect to their stable states at 298.15 K and 100 000 Pa: *hcp* for Er, gas for H. The Gibbs energies used for the reference states for Er are those proposed by the Scientific Group Thermodata Europe database [34].

4.2. Gas phase

The only specie considered in the gas phase is the di-hydrogen H_2. Consequently, the model used for the gas phase was:

$$G^{gas} = {}^{\mathrm{ref}}G^{gas}_{H_2} + RT\ln(P_{H_2})$$

4.3. Solid phases

All the solid phases were modelled using the sublattice model [35]. As shown in Table 1, the solid solution α_{Er} has an *hcp* crystal structure. The tri-hydride γ-ErH_3 is trigonal, HoD_3 structure type. This trigonal structure is an *hcp* structure where all the tetrahedral sites are occupied by hydrogen and last third of hydrogen is located for 1/3 in triangular sites and for 2/3 in displaced octahedral sites [11,12,14]. Because this trigonal structure is very

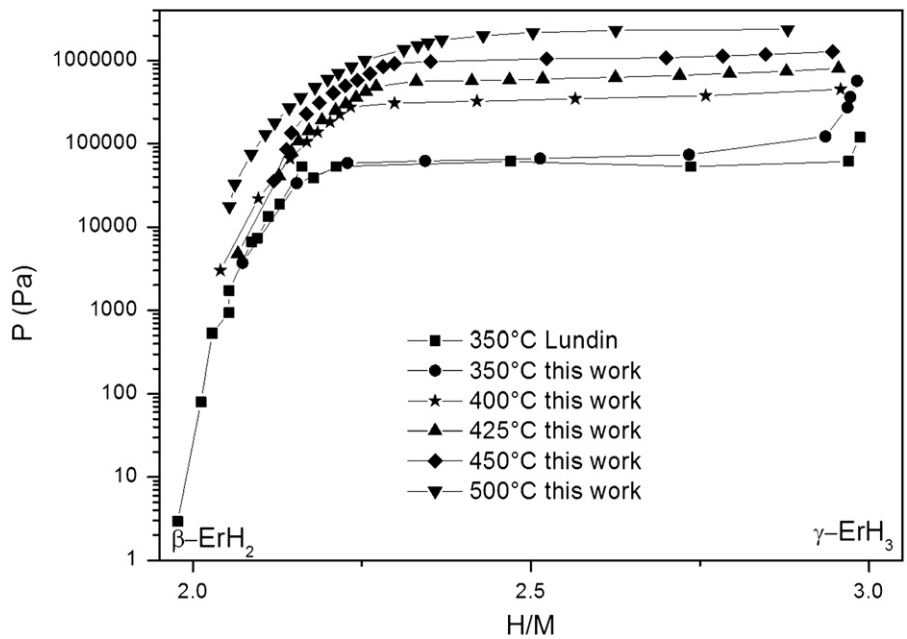

Fig. 5. PCT curves measured on the temperature range 350–500 °C between the two hydride phases (β-ErH_2 and γ-ErH_3) compared to Lundin's measurements at 350 °C [8].

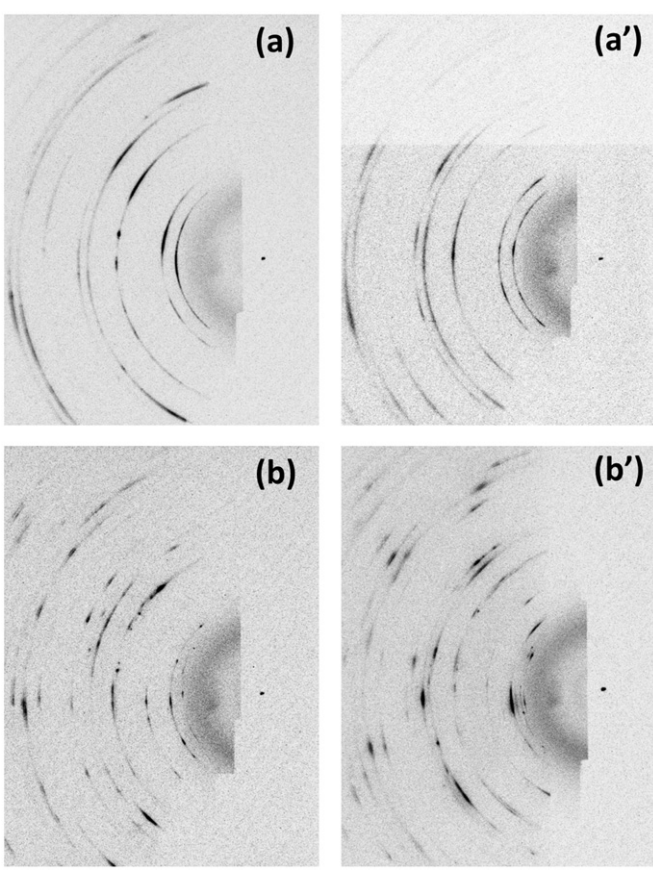

Fig. 6. μXRD patterns of β-ErH_2 (a) and (a'), and γ-ErH_3 (b) and (b') before and after ERDA+RBS analysis. On the left hand side: before the analysis and on the right hand side: after the analysis.

close to the *hcp* structure, it has been decided to model both phases with one unique model, which is a three-sublattice model. To model the *fcc* phase, β-ErH_2 structure type CaF_2, which can be hypostoichiometric and hyperstoichiometric, a three-sublattice model was also needed. So, the model chosen for all the solid phases, *hcp* and *fcc*, of this binary system was $(Er)(H,Va)^{O}(H,Va)_2^{T}$ with O referring to the octahedral (or triangular) sites and T to the tetrahedral sites. The Gibbs energy is described as the sum of three terms: reference, ideal, and excess.

$$G^{\varphi} = {}^{\mathrm{ref}}G^{\varphi} + {}^{\mathrm{id}}G^{\varphi} + {}^{\mathrm{ex}}G^{\varphi} \tag{1}$$

where:

$${}^{\mathrm{ref}}G^{\varphi} = y'_{\mathrm{Va}}y''_{\mathrm{Va}}G^{\varphi}_{\mathrm{Er:Va:Va}} + y'_{\mathrm{H}}y''_{\mathrm{Va}}G^{\varphi}_{\mathrm{Er}:H:\mathrm{Va}} + y'_{\mathrm{Va}}y''_{\mathrm{H}}G^{\varphi}_{\mathrm{Er:Va}:H} + y'_{\mathrm{H}}y''_{\mathrm{H}}G^{\varphi}_{\mathrm{Er}:H:H} \tag{2}$$

$${}^{\mathrm{id}}G^{\varphi} = RT((y'_{\mathrm{H}}\ln y'_{\mathrm{H}} + y'_{\mathrm{Va}}\ln y'_{\mathrm{Va}}) + 2(y''_{\mathrm{H}}\ln y''_{\mathrm{H}} + y''_{\mathrm{Va}}\ln y''_{\mathrm{Va}})) \tag{3}$$

$${}^{\mathrm{ex}}G^{\varphi} = y'_{\mathrm{H}}y'_{\mathrm{Va}}y''_{\mathrm{Va}}L^{\varphi}_{\mathrm{Er}:H,\mathrm{Va:Va}} + y'_{\mathrm{H}}y'_{\mathrm{Va}}y''_{\mathrm{H}}L^{\varphi}_{\mathrm{Er}:H,\mathrm{Va}:H} + y'_{\mathrm{H}}y''_{\mathrm{H}}y''_{\mathrm{Va}}L^{\varphi}_{\mathrm{Er}:H:H,\mathrm{Va}} + y'_{\mathrm{Va}}y''_{\mathrm{H}}y''_{\mathrm{Va}}L^{\varphi}_{\mathrm{Er:Va}:H,\mathrm{Va}} \tag{4}$$

In Eqs. (2)–(4), y' refers to site fraction in the second sublattice, y'' in the third and $L^{\varphi}_{A,\ \mathrm{B}}$ is the intseraction parameter between the atoms A and B. Usually, the interaction parameter $L^{\varphi}_{A,\ \mathrm{B}}$ is described by a Redlich–Kister [36] polynomial:

$$L^{\varphi}_{A,\ \mathrm{B}} = \sum_{v} {}^{v}L^{\varphi}_{A,\ \mathrm{B}}(x^{\varphi}_{\mathrm{A}} - x^{\varphi}_{\mathrm{B}})^{v}$$

The parameters $L^{\varphi}_{A,\ \mathrm{B}}$ can be temperature dependent:

$$L^{\varphi}_{A,\ \mathrm{B}} = a^{v,\ \varphi}_{A,\ \mathrm{B}} + b^{v,\ \varphi}_{A,\ \mathrm{B}} \times T + \cdots$$

For each end-member, the enthalpy of formation used, has been taken from the recent ab-initio calculations made by Joubert [24] and they were not optimized. The entropies of formation were taken from Beavis [18], in the case of the stable phases (α_{Er} as *hcp*, β-ErH_2 as *fcc* and γ-ErH_3 as *hcp*). For the other end-members, the value taken as entropy of formation was considered to be ideal. This ideal value is half the entropy of formation of the di-hydrogen molecule. So, the value of 65 J mol^{-1} per hydrogen atom was used and fixed for the entropy of formation. The entropy part of the lattice stability extrapolated by Cacciamani [25], for the erbium as pure *fcc*, has been used and also fixed.

Table 4
List of Gibbs energies and optimized parameters in J mol^{-1}.

Phase	Gibbs energies	Optimized parameters
hcp	$G^{Hcp}_{Er:Va:Va} = G^{SER}_{Er}$	${}^{0}L_{Er:*:H,va} = -31401 + 5.18*T$
	$G^{Hcp}_{Er:H:Va} = G^{SER}_{Er} + G^{SER}_{H} - 73{,}000 + 65 \times T$	
	$G^{Hcp}_{Er:Va:H} = G^{SER}_{Er} + 2*G^{SER}_{H} - 154{,}584 + 130 \times T$	
	$G^{Hcp}_{Er:H:H} = G^{SER}_{Er} + 3*G^{SER}_{H} - 256{,}006 + 210.35 \times T$	
fcc	$G^{Fcc}_{Er:Va:Va} = G^{SER}_{Er} + 2370 - 0.83 \times T$	${}^{0}L_{Er:H,va:*} = -20425$
	$G^{Fcc}_{Er:H:Va} = G^{SER}_{Er} + G^{SER}_{H} - 65{,}871 + 65 \times T$	${}^{0}L_{Er:*:H,va} = 45837 - 19.49*T$
	$G^{Fcc}_{Er:Va:H} = G^{SER}_{Er} + 2 \times G^{SER}_{H} - 210{,}001 + 147.35 \times T$	
	$G^{Fcc}_{Er:H:H} = G^{SER}_{Er} + 3 \times G^{SER}_{H} - 231{,}708 + 195 \times T$	

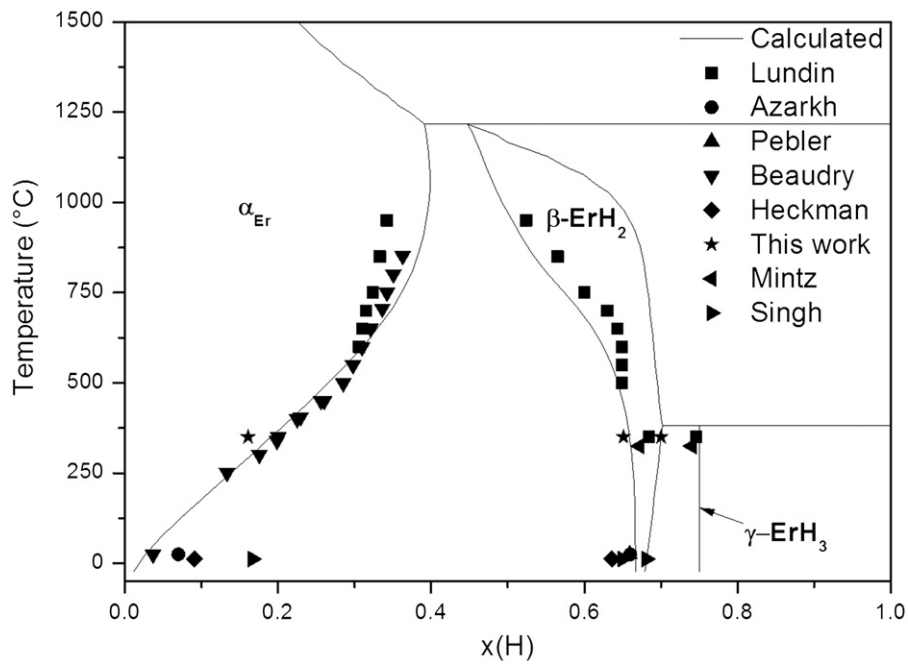

Fig. 7. The calculated Er–H binary phase diagram compared with some experimental measurements used during the modelling.

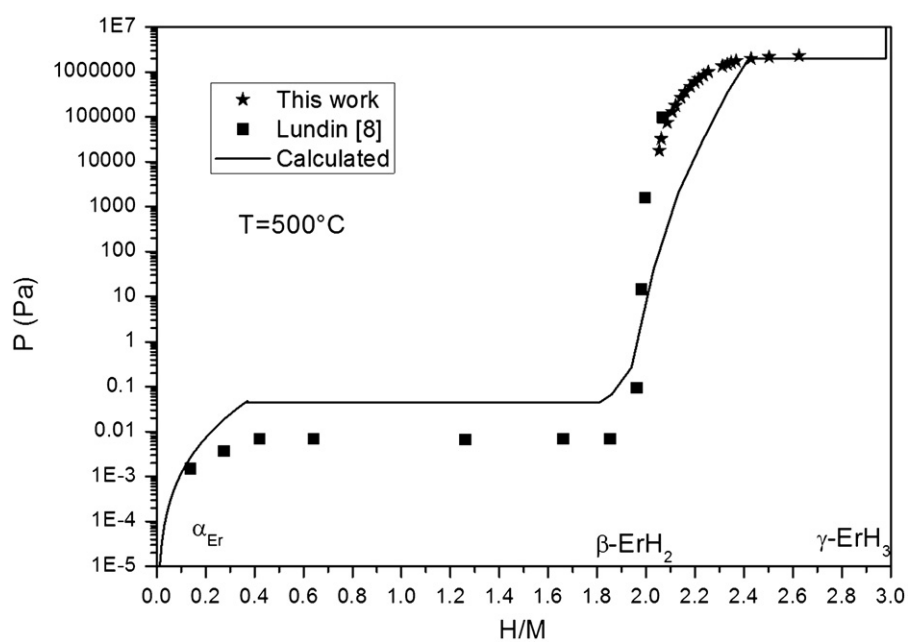

Fig. 8. PCT curve at 500 °C presenting the calculated curve and experimental data.

4.4. Data selection

All the data available on the Er–H binary system can be found in Table 2. The last column of this Table 2 is presenting whether the data has been used or not for the modelling.

Concerning the $\alpha_{Er}+\beta$-ErH_2 two phase region, Singh and Surplice data [16,17] were rejected from this optimization. In fact, they both showed very high hydrogen content on the solid solution α_{Er} (respectively 0.2 H/M and 0.23 H/M), far away from the data of Heckman also at 12 °C [15] or Pebler and Azarkh at 25 °C [6,7]. These data are also far from the curve designed by Lundin and Beaudry measurements [8,21]. Some data on the solubility limits obtained in our work were not consistent with the literature as presented in Section 3.5 and they were not used for the modelling. All the data, PCT curves measured between α_{Er} and β-ErH_2, α_{Er} solubility branches from Lundin and the ones from our work were taken into account except the α_{Er} solubility branch at 900 °C measured in this work.

On the β-$ErH_2+\gamma$-ErH_3 two phase region, all of the solubility limits have been used for the modelling except the measurements made by Surplice and from our work. In both cases, the data were not consistent with the other measurements. The PCT curves

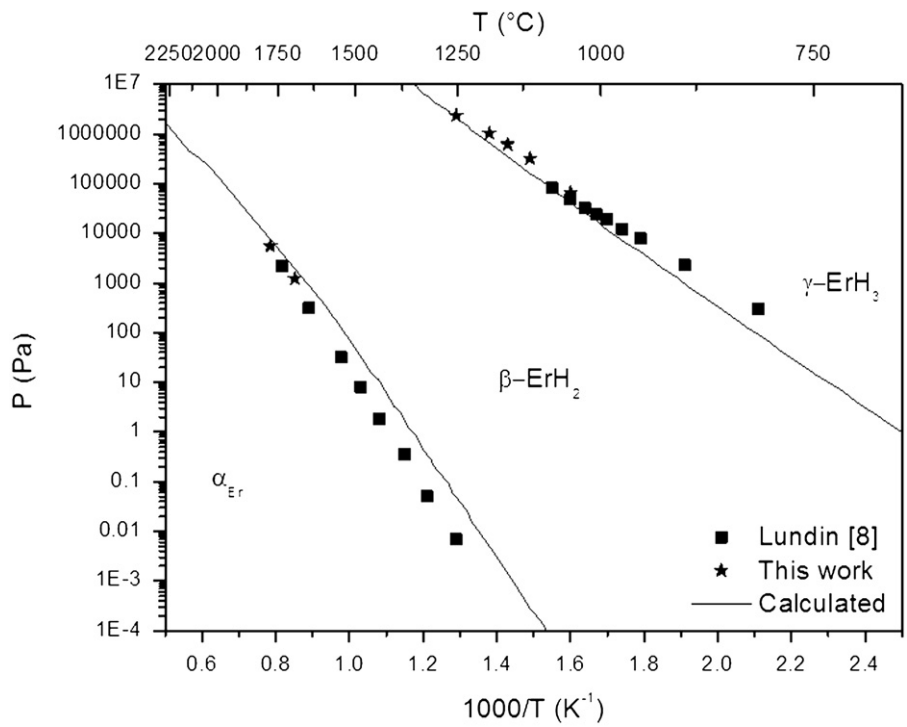

Fig. 9. Van't Hoff diagram showing the pressure and temperature couple for the equilibria between hydride phases.

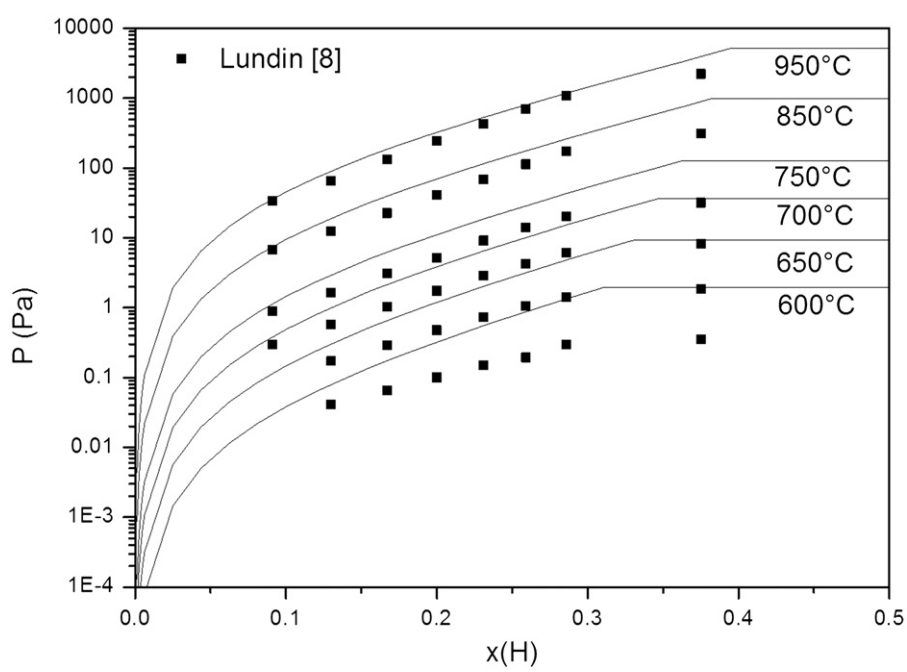

Fig. 10. Solubility branches of the α_{Er} phase in the temperature range 600–950 °C.

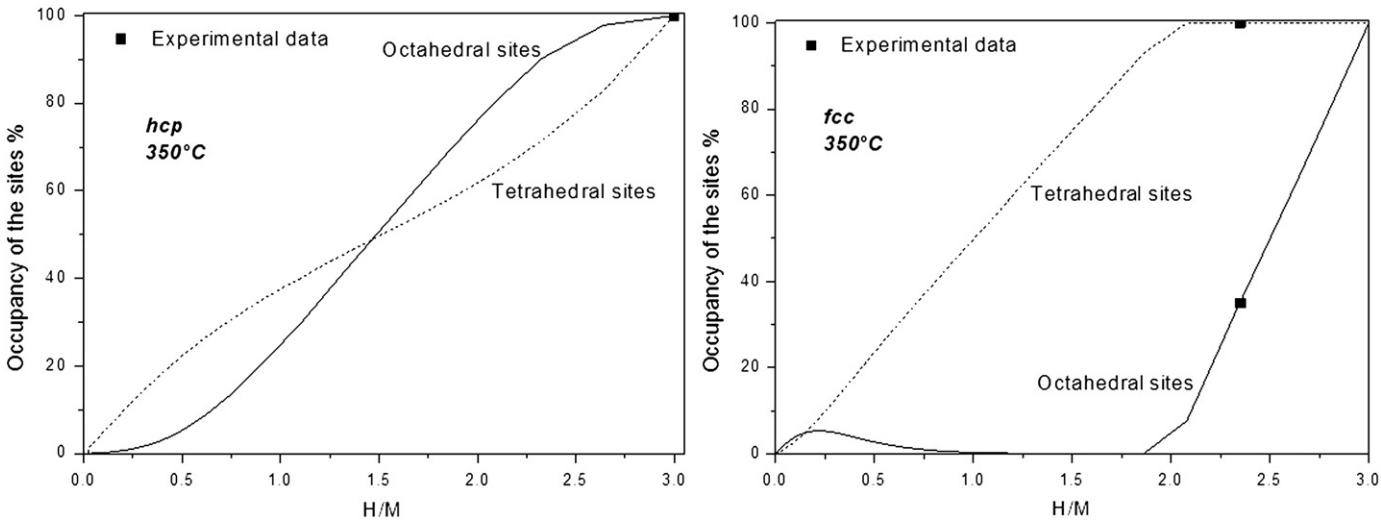

Fig. 11. Evolution of the occupancy of the interstitial sites as a function of the H/M composition in the *hcp* and *fcc* structures at 350 °C, compared with some experimental data [13].

measured on this domain by Lundin [8] were taken into account as the PCT curves measurements from our work. All the PCT curves of both two phase region are shown as plateau pressure in Table 2.

4.5. Results

The optimization has been carried out using the Parrot module of the Thermo-Calc software [33]. The five optimized parameters are given in Table 4, and the calculated Er–H phase diagram is presented in Fig. 7 and compared with some experimental data used for the modelling. This binary system shows a high solubility of hydrogen in the solid solution α_{Er}. The β-ErH_2 phase presents a large extension at low hydrogen content and at high temperature. The γ-ErH_3 phase has a small composition domain. At last, two invariant reactions can be found (for $P=10^5$ Pa):

- Gas-peritectoiod: Gas (H_2)+β-ErH_2→γ-ErH_3 at 433 °C
- Gas-peritectoiod: Gas (H_2)+α_{Er}→β-ErH_2 at 1161 °C

4.6. Discussion

In order to check the quality of the modelling of the Er–H binary system, graphs are presented with the calculated curves compared with experimental measurements.

Fig. 8 presents a PCT curve calculated at 500 °C compared to the experimental values taken from Lundin [8] or measured during this work. This calculated PCT curve is coherent with the experimental data. This curve is presenting two plateaus, each one corresponding to a two phase region, the first one from α_{Er} to β-ErH_2 and the second one from β-ErH_2 to γ-ErH_3.

A Van't Hoff diagram showing the formation couple (*P* and *T*) of the two hydrides is presented in Fig. 9: above the curve, the hydride exists. This calculated Van't Hoff diagram is in good agreement with the experimental values.

Fig. 10 shows the solubility branches of the α_{Er} phase for several temperatures (from 600 °C to 950 °C). For each isotherm, the calculated solubility branch fits well the experimental measurement of Lundin [8].

The hydrogen occupancy in the different interstitial sites has been calculated in the *hcp* and *fcc* cases. It is shown in Fig. 11 and compared with some experimental data [13]. It can be noticed that whatever the structure is, at 350 °C, first the tetrahedral interstitial sites are filled and then octahedral interstitial sites. The higher stability of the hydrogen atoms in the tetrahedral sites have already been seen by Yang et al. [23].

Finally, in this optimisation, it has been chosen to optimise only five parameters. This limited number of parameters allows reproducing rather well the experimental data.

5. Conclusion

The erbium–hydrogen binary system has been studied experimentally. Different but complementary experimental techniques (XRD, PCT curves and the ERDA+RBS combination) have been used in order to find the equilibria and the phase compositions. A new evaluation of this system has been proposed. This binary system has been assessed using the Calphad method and the Parrot module of Thermoc-Calc. The sublattice model has been chosen for all the solid phases. The Er–H binary phase diagram is in rather good agreement with the experimental data. This optimisation has been made with only five parameters. Finally, this system shows a larger extension of the α_{Er} solid solution and the hydride β-ErH_2 phase at high temperature than reported before.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.calphad.2013.01.004.

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